CASE STUDY · FULL-STACK WEB APPLICATION

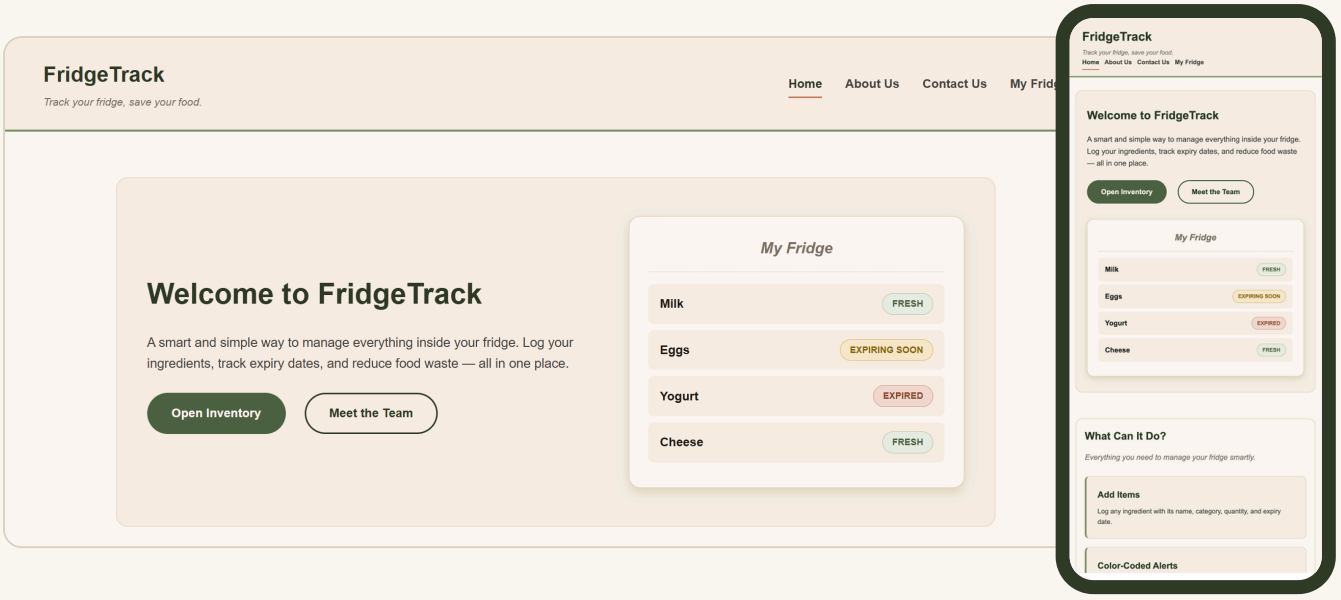
FridgeTrack

Track Your Fridge, Save Your Food

🏆 ACHIEVEMENT

Best Web Development Project

Elite Best Project Award · Software Engineering Dept., University of Jeddah



A full-stack web application designed to help users track food items, quantities, categories, and expiry dates to reduce food waste.

MY ROLE	TEAM	COURSE	YEAR
Front-End & UI Design	**5 students**	**CCSW-321 Web Development**	**2025–2026**

01 · THE PROBLEM

Food gets forgotten at the back of the fridge

Without a clear view of what is inside the fridge and when it expires, items are forgotten until they spoil, and groceries are bought twice.

60% of the world's wasted food comes from **households**.

Source: UNEP Food Waste Index Report 2024

No visibility

Users don't remember what they have or how much is left.

Missed expiry dates

Dates are printed on packaging, not tracked anywhere useful.

Duplicate buying

The same items are bought again while old ones go to waste.

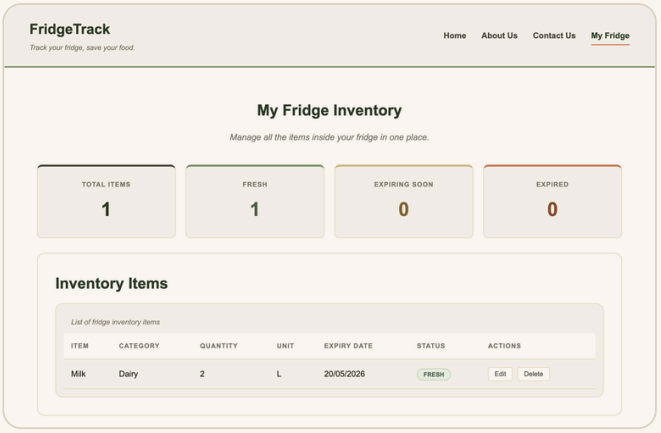
02 · THE SOLUTION

One simple inventory with color-coded expiry

Users log what they store, and FridgeTrack shows at a glance what is fresh, what to use soon, and what has expired.

KEY FEATURES

- Add / Edit / Delete food items
- Automatic expiry status
  SAFE · EXPIRING SOON · EXPIRED
- Categories, quantities & units
- Dashboard summary cards
- Input validation & sanitization



Summary cards and the live inventory table

HOW IT WORKS

STEP 01
Open the inventory
Go to "My Fridge" to see every stored item.

STEP 02
Add your items
Enter name, category, quantity, unit and expiry date.

STEP 03
Track & manage
Statuses update by color; edit or delete anytime.

03 · METHODOLOGY

How we built it

01 — Plan
Defined the idea, core features and the four site pages.

02 — Design
Built a calm visual identity with colors that carry meaning.

03 — Build
Developed the responsive front-end and connected it to the API and database.

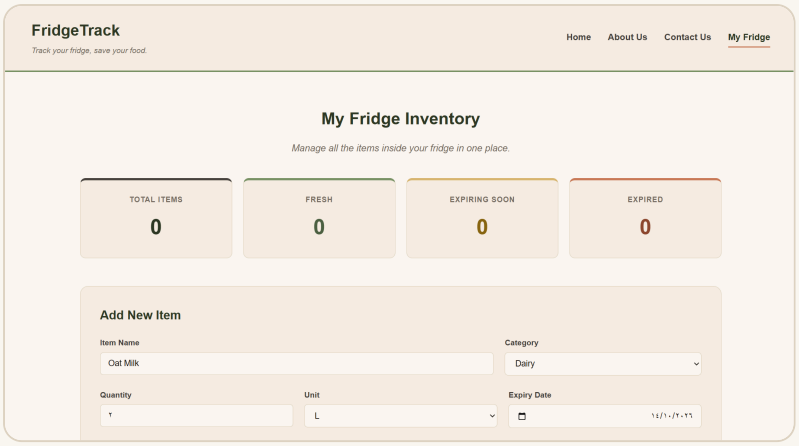
04 — Test
Validated HTML, checked screen sizes and tested all forms.

Architecture: Browser · HTML, CSS, JS → Node.js + Express API → MySQL Database

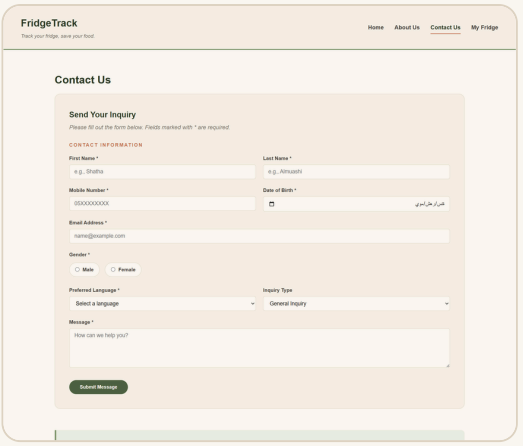
HTML · CSS · JavaScript · Node.js · Express · MySQL · Git / GitHub

04 · INTERFACES

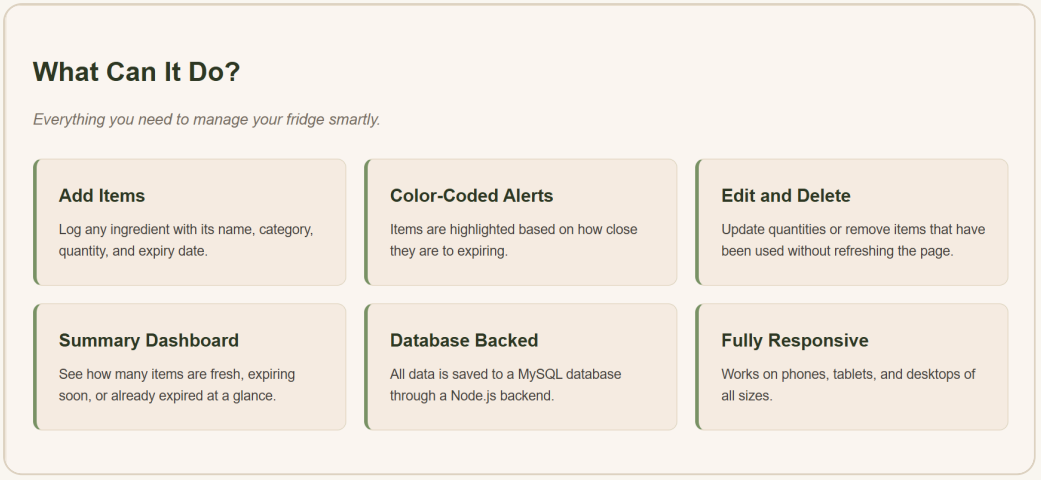
Designed to be understood at a glance



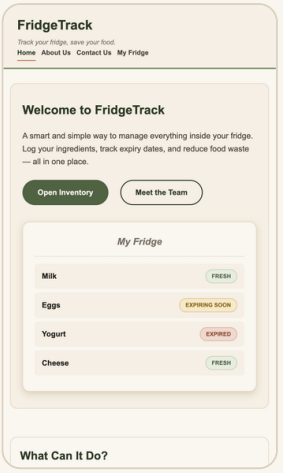
My Fridge: summary cards + Add New Item form



Contact Us: validated form



Home: feature overview



Mobile view

05 · MY CONTRIBUTION

Front-End Developer & UI Designer

- Built responsive pages with HTML, CSS & JavaScript
- Designed the UI: color palette, layout & status badges
- Implemented client-side form validation
- Made the site work across phone, tablet & desktop
- Connected the interface to the back-end API

06 · ACHIEVEMENT

Best Web Development Project

Won the Elite Best Project Award, organized by the Software Engineering Department at the University of Jeddah.